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# AUTO TICKET CLASSIFICATION USING FLOW DESIGNER.

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3rd – B.SC INFORMATION TECHNOLOGY

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# **Auto Ticket Classification using Flow Designer**

## **Problem Statement**

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers.

Currently, IT staff manually reviews each request and assigns a category, which is time-consuming and inefficient.

This project automates ticket classification using ServiceNow Flow Designer by analyzing keywords in the incident's short description and description fields.

## **Objective**

- Automatically classify incidents at the time of creation.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.

## **PHASE 1: Requirement Analysis & Planning**

The school IT helpdesk receives multiple support requests daily from students and teachers related to common issues such as network connectivity, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket to:

- Identify the issue type.
- Select the correct category and subcategory.
- Inform the caller about ticket creation.

This manual process is time-consuming, error-prone, and not scalable as the number of requests increases.

**Activity : 1 Business Use Case.**

## Business Requirements

The system must:

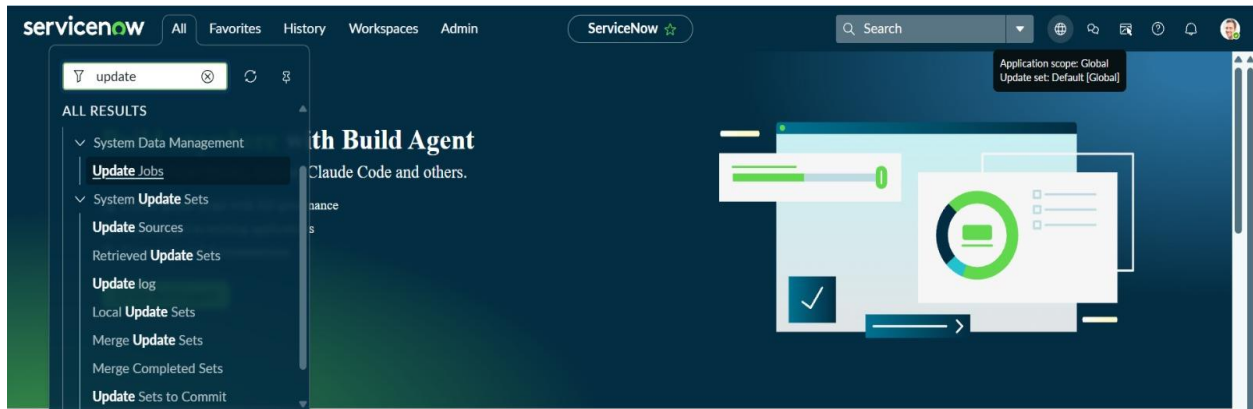
1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

## Activity : 2 Creation of New Update Set.

### Navigation Steps :

- Go to All menus → Search for Update set.
- Under System Update Sets → Select Local Update Sets.
- Click the New button on the top right corner of the page.
- Name : **Project Update Set.**
- State : **In progress.**
- Application : **Global.**
- Then **Save** the **Form.**
- Under Related links Click on the **Make This My Current Set.**
- Then the Created Update Set will appear as Current.

## Activity : 3 Navigation steps in Picture View.



**Update Sets** | Name | Search | Actions on selected rows... | **New**

| Name    | Application | State       | Installed from | Created             | Created by | Parent  | Batch Base |
|---------|-------------|-------------|----------------|---------------------|------------|---------|------------|
| Default | Global      | In progress |                | 2026-01-06 03:50:56 | system     | (empty) | (empty)    |

Related Links  
[Merge Update Sets](#)

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**Update Set**  
New record

\* Name  Application

State

Parent

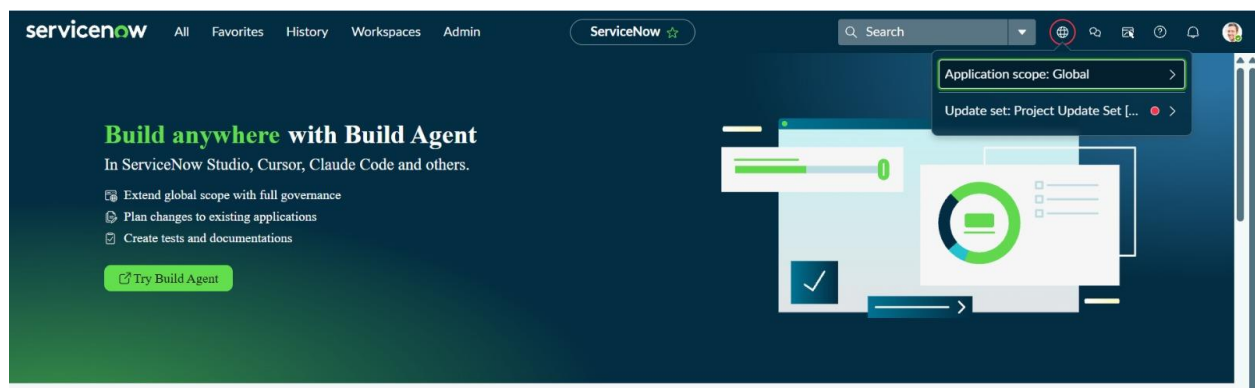
Release date

Description

The screenshot shows the 'Update Set' form in ServiceNow. The form is titled 'Update Set' and 'Project Update Set'. It contains several fields:
 

- Name:** Project Update Set (highlighted with a red box)
- State:** In progress (dropdown menu)
- Parent:** (empty field with a search icon)
- Release date:** (calendar icon)
- Install date:** (empty field)
- Installed from:** (empty field)
- Description:** (text area with a code icon)
- Application:** Global (dropdown menu)
- Created:** 2026-02-06 20:55:31
- Created by:** admin
- Merged to:** (empty field)

 Below the form, there are 'Update' and 'Delete' buttons. A 'Related Links' section is also visible, containing links like 'Make This My Current Set', 'Merge With Another Update Set', 'Scan Update Set', and '(SN Utils) Versions (0)'. At the bottom, there is a table with columns: Created, Type, View, Target name, Updated by, Remote update set, and Action. The table currently shows one row with the value 'Project Update Set' in the 'Created' column.



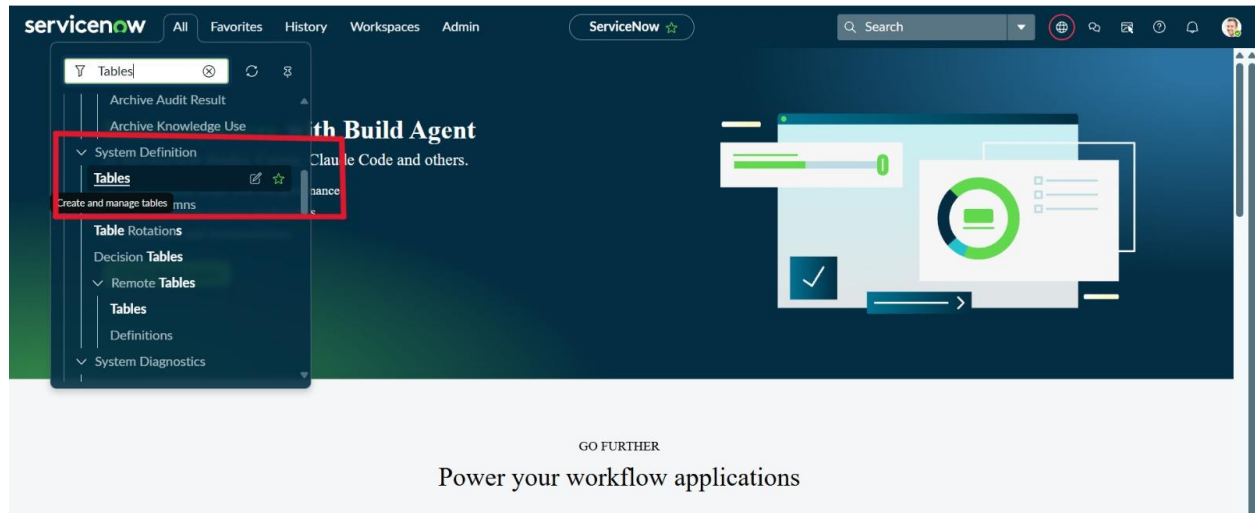
## Phase :- 2 Backend Development & Configuration

### Activity 1 : Creation of Custom Table to Store Ticket Records.

#### Steps to Create Custom Table

- Open All menus → Search for Tables.
- Under **System Definition** → Click **Tables**.
- On the top right corner Click **New** button.
- Label : **Incident WorkFlow**.
- Uncheck the **Create module**.

- Now **Save** the Form.
- Then Click the **Controls** Suction and Enable the Check box **true Auto Number**.
- Now again **save** the Form.



| Label                           | Name                       | Extends table    | Extensible | Updated             |
|---------------------------------|----------------------------|------------------|------------|---------------------|
| Search                          | Search                     | Search           | Search     | Search              |
| Logged in User                  | v_user_session             | (empty)          | false      | 2007-07-13 16:15:23 |
| System Plugin                   | v_plugin                   | (empty)          | false      | 2007-10-09 12:10:05 |
| Active Transactions             | v_transaction              | (empty)          | false      | 2008-05-23 17:30:17 |
| Node Log                        | v_logfiles                 | (empty)          | false      | 2009-05-14 17:05:41 |
| Customer Uploads                | v_customer_uploads         | (empty)          | false      | 2009-05-14 18:38:13 |
| MySQL Process List              | v_mysql_proclist           | (empty)          | false      | 2012-02-25 19:47:02 |
| MySQL Status                    | v_mysql_status             | (empty)          | false      | 2012-02-25 19:47:08 |
| MySQL Variables                 | v_mysql_variables          | (empty)          | false      | 2012-02-25 19:47:12 |
| IOStats                         | v_iostats                  | (empty)          | false      | 2012-02-27 17:13:51 |
| Cluster Nodes                   | v_cluster_nodes            | (empty)          | false      | 2012-02-27 23:48:26 |
| SQL Debug                       | v_sql_debug                | (empty)          | false      | 2012-02-28 16:40:55 |
| Workflow Validation Report      | v_wf_validation_report     | (empty)          | false      | 2012-10-01 10:43:05 |
| Hyper-V Resource Pool Component | cmdb_ci_hyper_v_rpool_comp | Hyper-V Object   | false      | 2012-10-30 08:07:25 |
| Application Menu                | sys_app_application        | Application File | false      | 2012-10-30 13:59:18 |
| Menu Category                   | sys_app_category           | Application File | false      | 2012-10-30 13:59:44 |
| Scriptable Object               | v_scriptable_object        | (empty)          | false      | 2012-11-09 15:15:23 |

Table  
New record

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

\* Label Incident Workflow

\* Name u\_incident\_workflow

Extends table

Application Global

Create module

Remote Table

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

| Column label        | Type | Reference | Max length | Default value | Display |
|---------------------|------|-----------|------------|---------------|---------|
| Insert a new row... |      |           |            |               |         |

Submit Cancel

Related Links

Table  
Incident Workflow

Columns Controls Application Access

Extensible

Live feed

Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format

Auto-number

Prefix INC

Number 500

Number of digits 5

Security Rules (ACLs) are required if anyone other than an administrator needs to work with this table. Creating default security rules will grant full access to this table to anyone with the user role you specify.

Create access controls

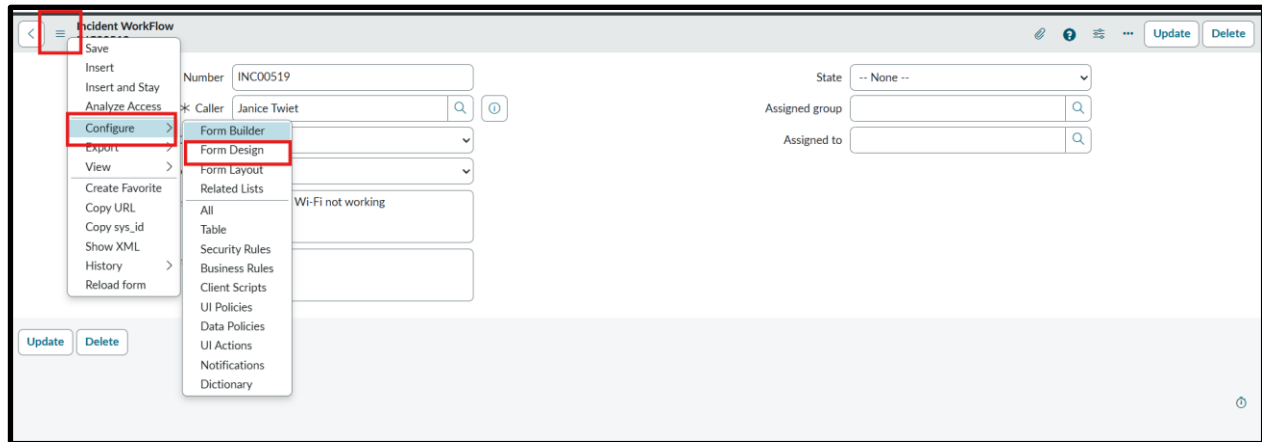
\* User role u\_incident\_workflow\_user

Delete Update Delete All Records

Related Links

Form Builder

- Then Save the Form and Under Related Links Click **Show Form**.
- Now Open the Configure On clicking the Context Menu and then Open the **Form Designer**.
- Then Create Fields as Required.

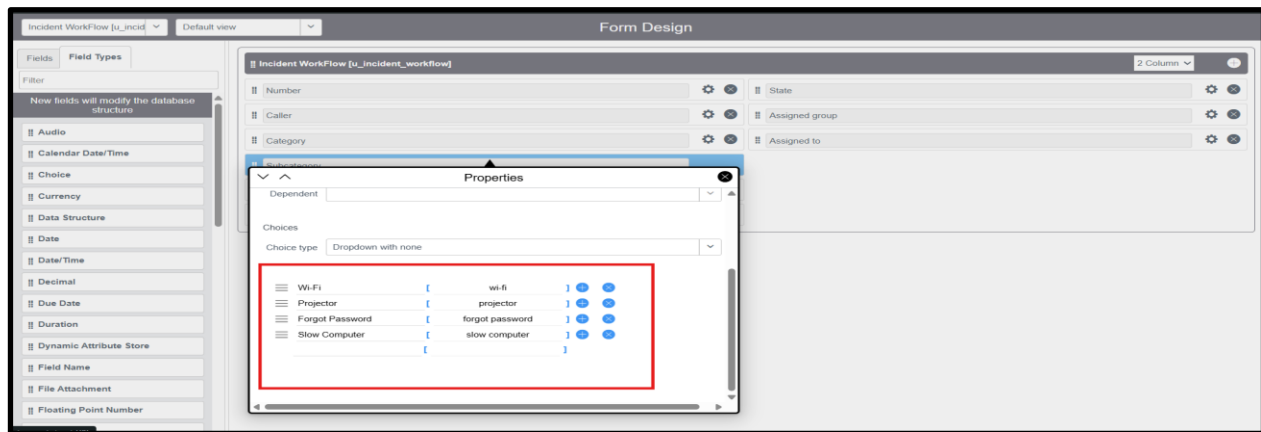
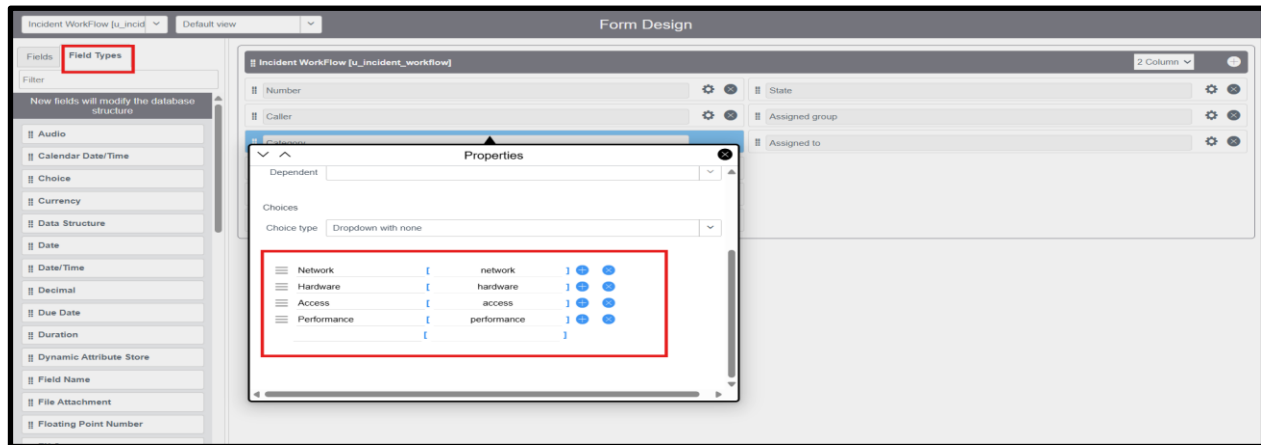


## Activity 2: Field Creation and Data Type Configuration

The objective of this milestone is to create all required fields in the School IT Ticket table with correct data types, choices, and dependencies, ensuring structured and consistent data capture.

| Field Label       | Type        | Reference/Choice                               |
|-------------------|-------------|------------------------------------------------|
| Number            | Auto Number | —                                              |
| Caller            | Reference   | sys_user                                       |
| Category          | Choice      | [Network,Hardware,Access,Performance]          |
| Subcategory       | Choice      | [Wi-Fi,Projector,Forgot Password,Slow Compter] |
| Short Description | String      | —                                              |
| Description       | String      | —                                              |
| State             | Choice      | [New,In progress,On hold,Resolved,Closed]      |
| Assigned Group    | Reference   | sys_user_group                                 |
| Assigned to       | Reference   | sys_user                                       |

- After Creating Fields with their appropriate Type, then add the Choice from the Form design.
- To all the Choice list Fields.
- They are Category,Subcategory and State.



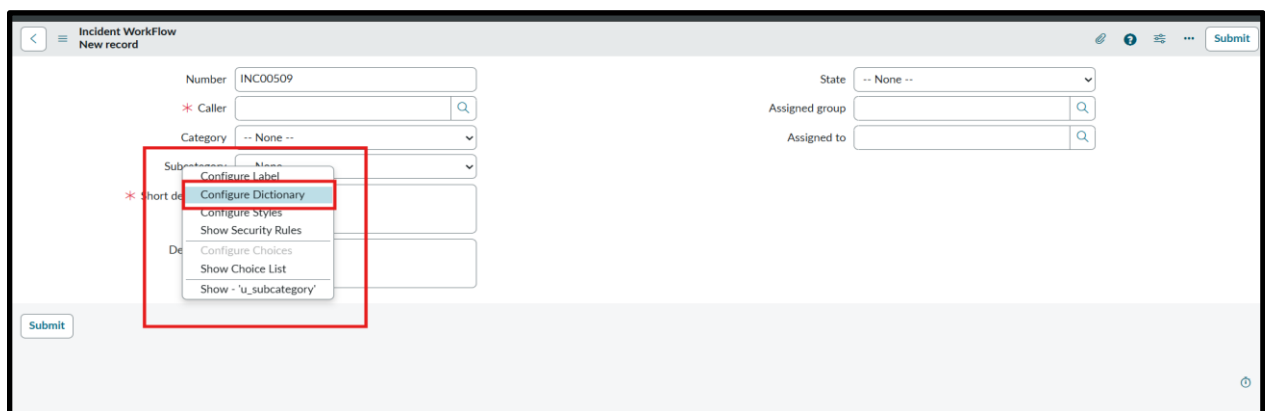
- Same like that, add Choice to the State field.
- Then Click the save button on the top right corner.
- Open the Form and reload it.

## Activity 3 : Implementing Dependency Between Category and Subcategory Choice Fields

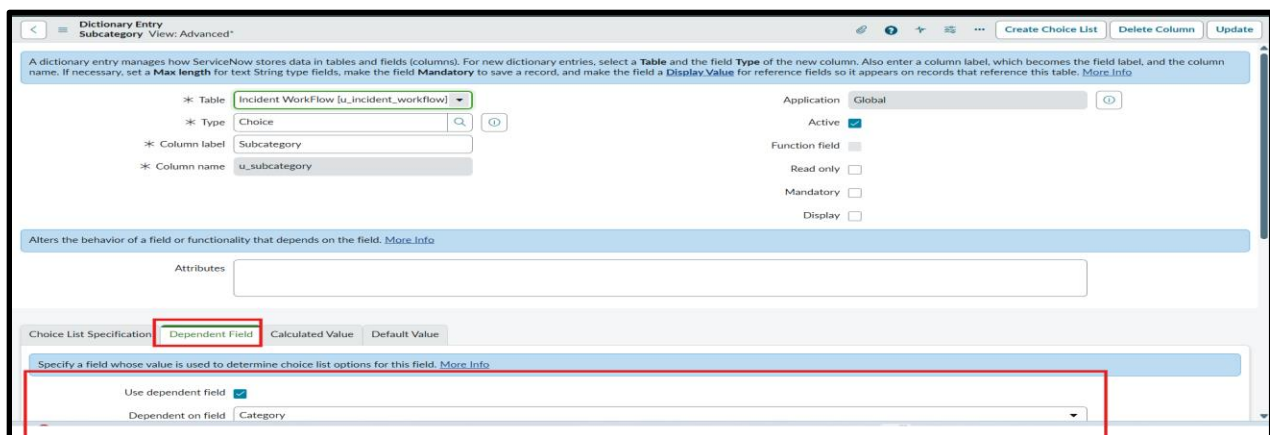
The objective of this milestone is to implement a **dependency relationship** between the **Category** and **Subcategory** fields so that the available Subcategory options dynamically change based on the selected Category.

### Navigation Steps :

- Open the Created Table Form and Right click on the **Subcategory** Field.
- Click on the **Configure Dictionary**.
- Under **Related links** Click on the **Advanced view**,
- Then select the **Dependent Field** Section.
- Enable the checkbox **Use dependent field true**.
- Then select the Field **Category** on the **Dependent on field**.
- Then **save** the Form.



This screenshot shows the 'Incident Workflow' form in 'New record' mode. The 'Subcategory' field is selected, and a context menu is open. The menu options are: Configure Label, Configure Dictionary (highlighted with a red box), Configure Styles, Show Security Rules, Configure Choices, Show Choice List, and Show 'u\_subcategory'. The 'Category' field is set to '-- None --'.



This screenshot shows the 'Dictionary Entry' configuration page for the 'Subcategory' field. The 'Table' is 'Incident Workflow [u\_incident\_workflow]' and the 'Type' is 'Choice'. The 'Column label' is 'Subcategory' and the 'Column name' is 'u\_subcategory'. The 'Application' is 'Global' and the 'Active' checkbox is checked. The 'Dependent Field' tab is selected, and the 'Use dependent field' checkbox is checked. The 'Dependent on field' is set to 'Category'.

- Now scroll down to the choice section.

- Then add the dependent fields for which option dependent on the category gives the category choice backend name to the subcategory choice values.
- Wi-Fi → Network.
- Projector → Hardware.
- Forgot password → Access.
- Slow computer → Performance.

Specify a field whose value is used to determine choice list options for this field. [More Info](#)

Use dependent field ☒

Dependent on field Category

[Create Choice List](#) [Delete Column](#) [Update](#)

Related Links  
[Show Table](#)  
[Run Point Scan](#)  
[\[SN Utils\] Versions \(0\)](#)

Access Controls Choices (4) Attributes Labels (1)

Label Search Actions on selected rows... New

| Label           | Value           | Language | Sequence | Inactive | Updated             | Dependent value |
|-----------------|-----------------|----------|----------|----------|---------------------|-----------------|
| Forgot Password | forgot password | en       | 2        | false    | 2026-02-06 21:45:02 | access          |
| Wi-Fi           | wi-fi           | en       | 0        | false    | 2026-02-06 21:44:50 | network         |
| Slow Computer   | slow computer   | en       | 3        | false    | 2026-02-06 21:45:15 | performance     |
| Projector       | projector       | en       | 1        | false    | 2026-02-06 21:45:36 | hardware        |

Insert a new row...

- Then save the Form.
- Then it will display each choice to each one separately based on the category subcategory displays.

Incident WorkFlow  
New record

Number

\* Caller

Category Network

Subcategory -- None --

\* Short description

Description

State -- None --

Assigned group

Assigned to

[Submit](#)

Incident WorkFlow

New record

Submit

Number

INC00509

\* Caller

Category

Hardware

Subcategory

-- None --

\* Short description

-- None --

Projector

Description

State

-- None --

Assigned group

Assigned to

Submit

Incident WorkFlow

New record

Submit

Number

INC00509

\* Caller

Category

Access

Subcategory

-- None --

\* Short description

-- None --

Forgot Password

Description

State

-- None --

Assigned group

Assigned to

Submit

Incident WorkFlow

New record

Submit

Number

INC00509

\* Caller

Category

Performance

Subcategory

-- None --

\* Short description

-- None --

Slow Computer

Description

State

-- None --

Assigned group

Assigned to

Submit

### Milestone 3 : Automation using Flow Designer & Email Notification.

When a student creates a new IT support ticket, they only need to enter the Caller Name and Short Description of the issue. The student does not need to manually select the Category or Subcategory. Once the ticket is submitted, a Flow Designer automation is triggered and evaluates the Short Description for predefined keywords such as WiFi, Projector, Password, or Slow. Based on the identified keyword, the flow automatically assigns the appropriate Category and Subcategory to the ticket.

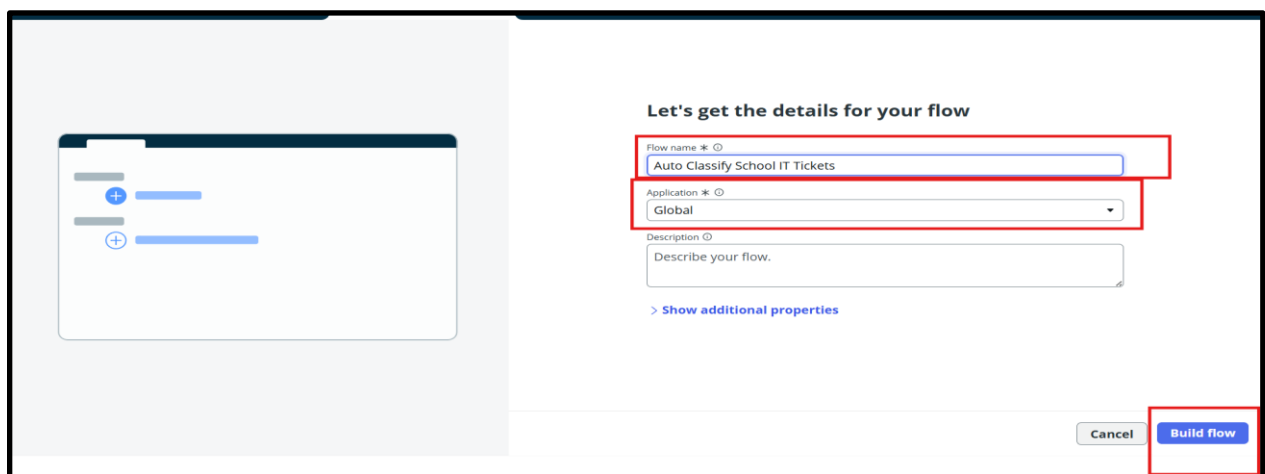
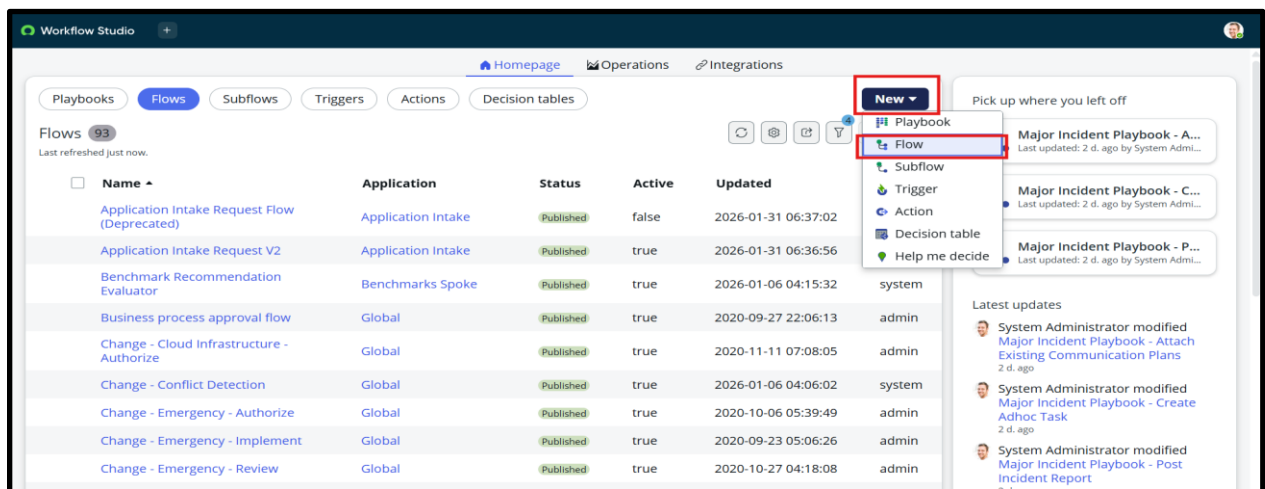
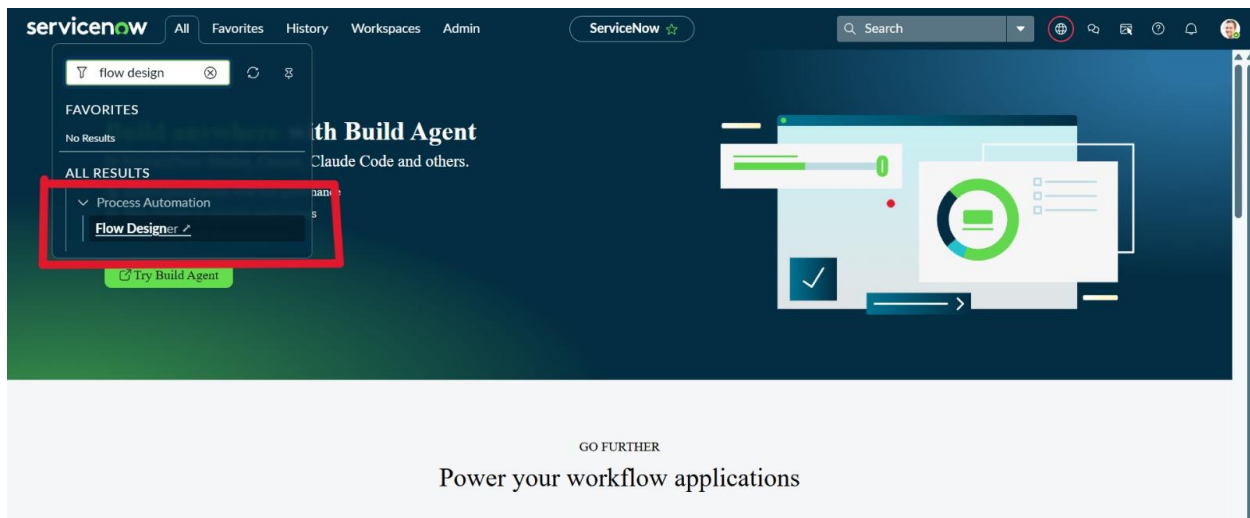
- If the short description contains the word *WiFi*, the flow assigns **Category = Network** and **Subcategory = WiFi**.
- If the description contains *Projector*, the flow assigns **Category = Hardware** and **Subcategory = Projector**.
- If the description includes *Password* or *Login*, the flow assigns **Category = Access** and **Subcategory = Forgot Password**.
- If the description mentions *Slow* or *Hanging*, the flow assigns **Category = Performance** and **Subcategory = Slow Computer**.

This automation ensures consistent classification, eliminates human error, and significantly reduces the manual effort required by IT staff.

The entire process is implemented using **Flow Designer**, making the solution easy to maintain, scalable, and completely no-code.

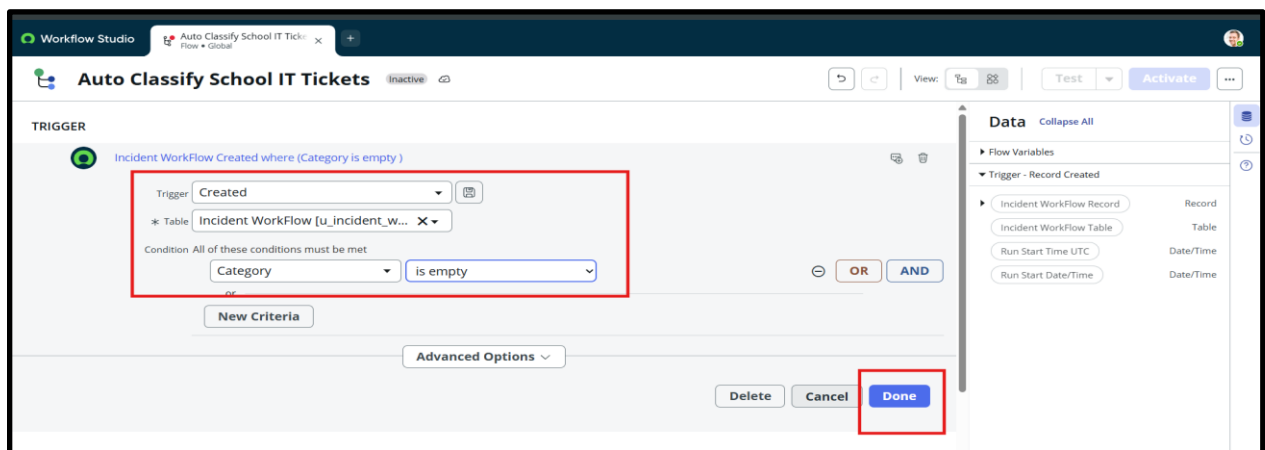
#### Activity : 1 Auto Classification using Flow design.

- Go to **All menus** → Search **Flow designer**.
- Under **Process Automation** Open the **Flow designer**.
- On the top right corner Click **New** and Select **Flow** Option.
- Name : **Auto Classify School IT Tickets**.
- Application : **Global**.
- Now Click the **Build Flow** button.



## Activity : 2 Trigger Configuration.

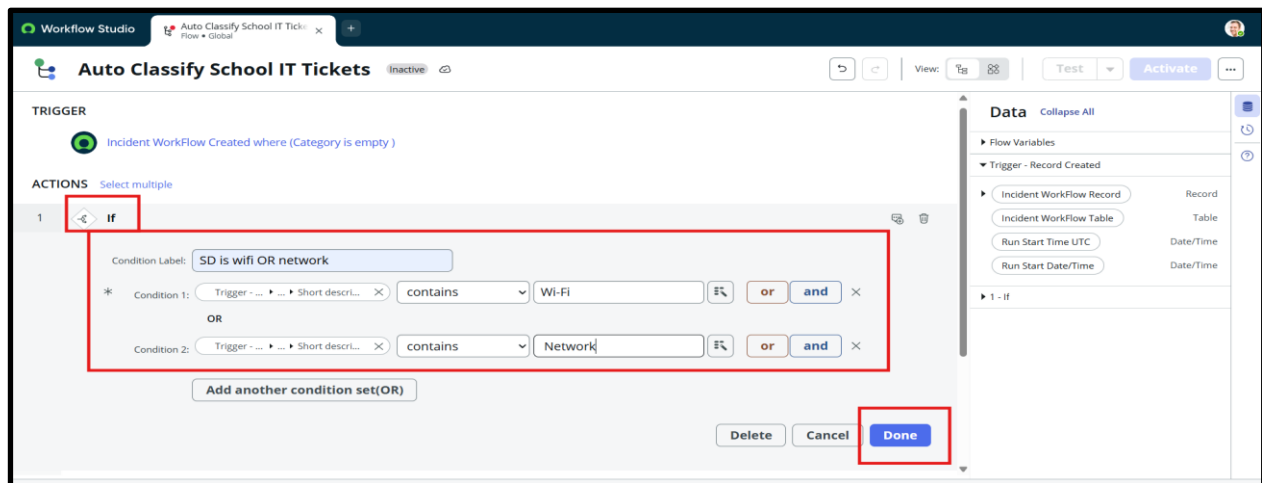
- Trigger : **Record Created.**
- Table : **Incident Workflow.**
- Condition : **Category is Empty.**
- Then Click **Done.**



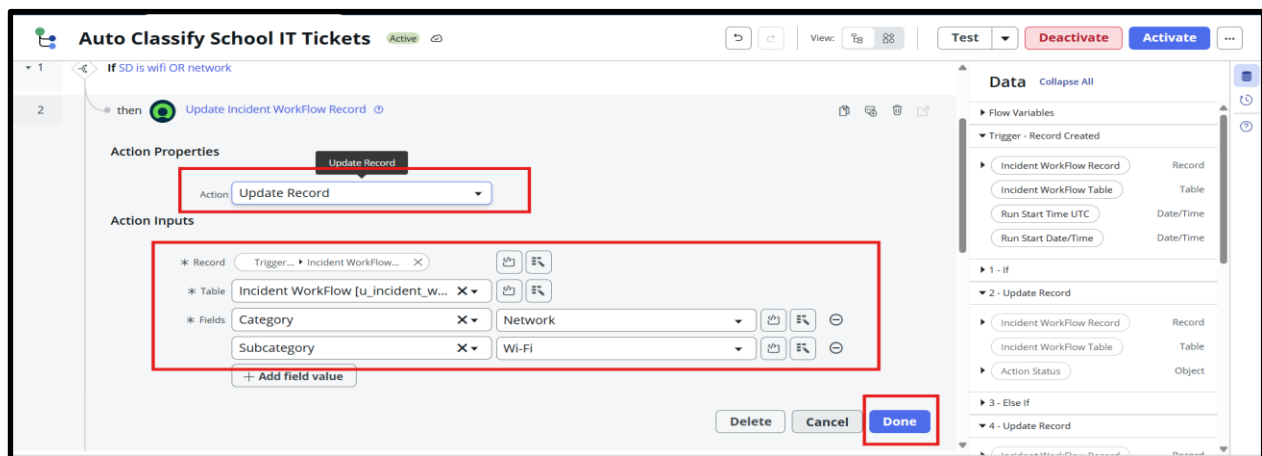
- Now Under **Actions** Select **Flow Logic** Option.

## Activity : 3 Wi-Fi Issue Classification.

- Select **If** condition.
- Condition label : SD is **Wifi or Network** (Give as ur wish).
- Condition 1 : Using data pill picker Select **Short description** as Field , and operator **Contains** , Value is **Wi-Fi**.
- Condition 2 : Using data pill picker Select **Short description** as Field, and Operator is **Contains**, Value is **Network**.
- Then Click **Done** Button.



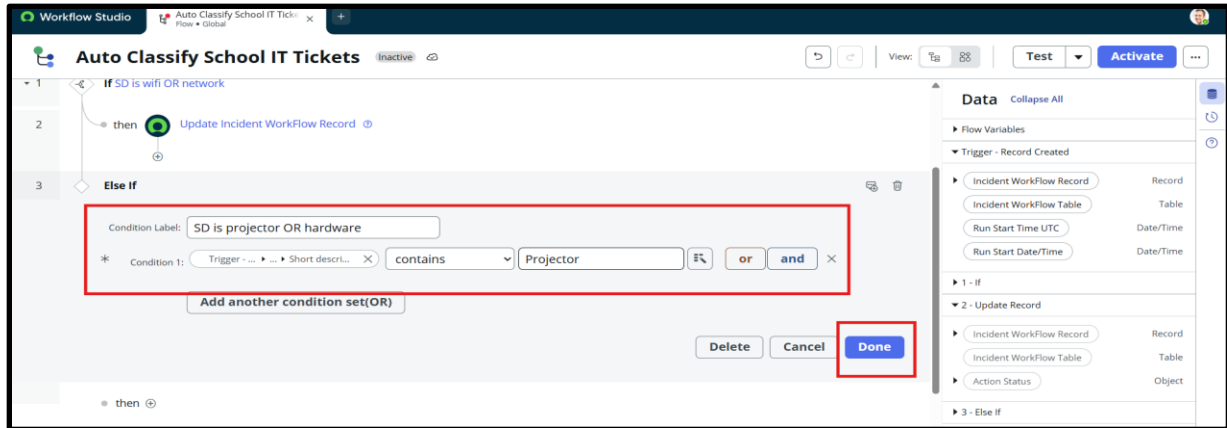
- Now Click **Action to Update record.**
- Action Inputs.
- In Record Field drag and drop data from the right side.
- Table : Auto updates as Incident Workflow.
- Fields Set **Category = Network.**
- Set **Subcategory = Wi-Fi.**
- Then Click the **Done** button.



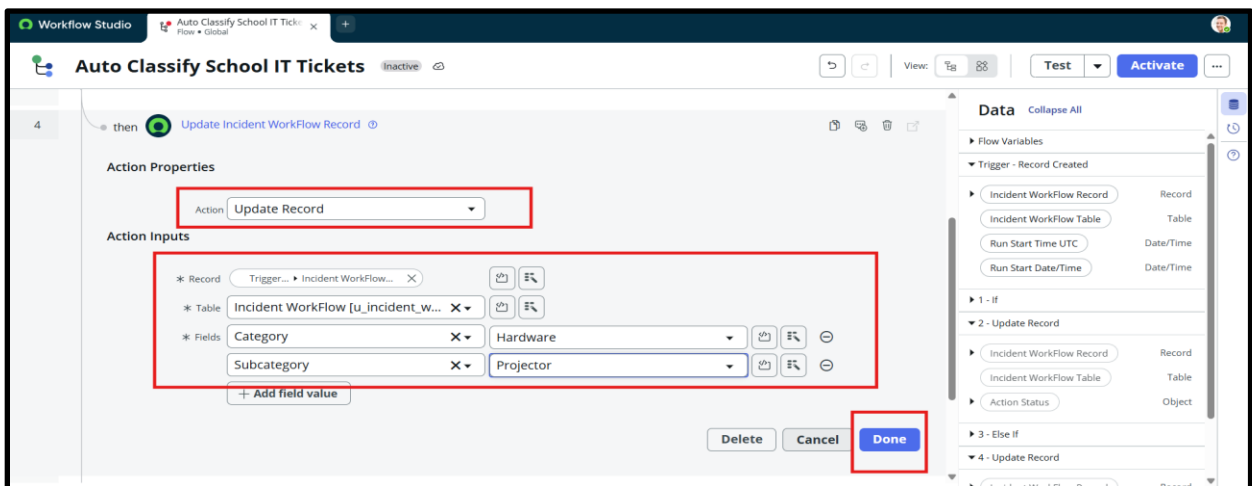
## Activity : 4 Projector Issue Classification.

- Now Select the **Else If Condition.**
- Condition label : SD is **Projector or Hardware.**
- Condition : **Short description Contains Prajector.**

- Then Click the **Done** button.
- Now Action is an **Update record**.
- Set **Category** = **Hardware**.



- Set **Subcategory** = **Projector**.

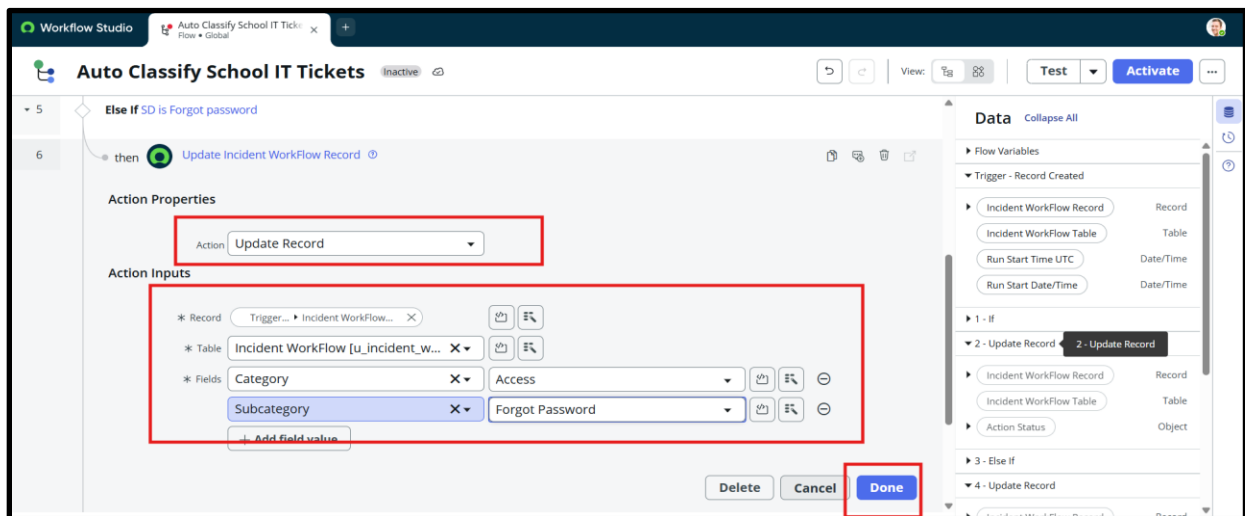
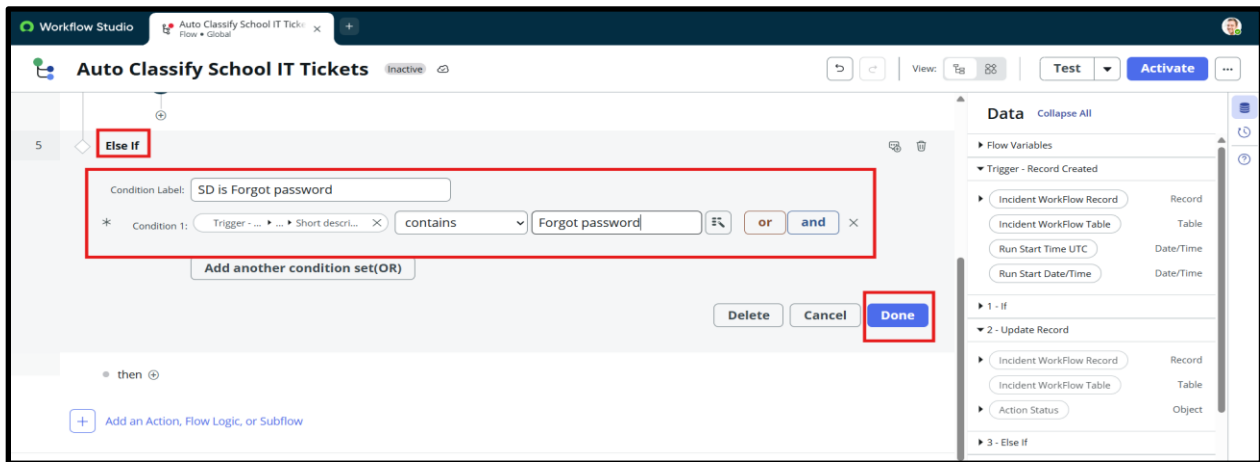


- Then Click the **Done** button.

## Activity : 5 Password Issue Classification.

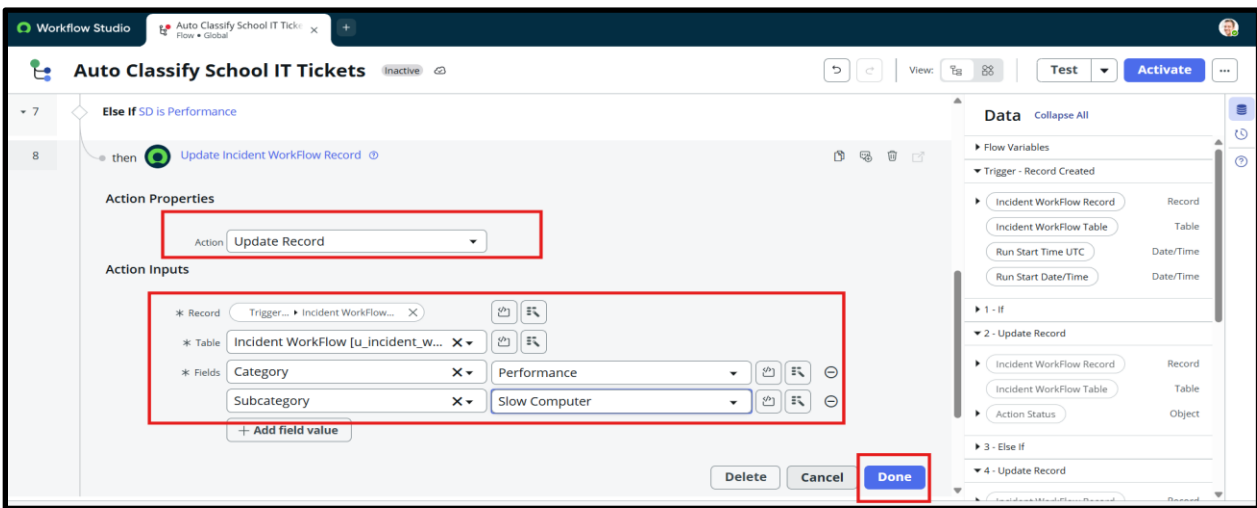
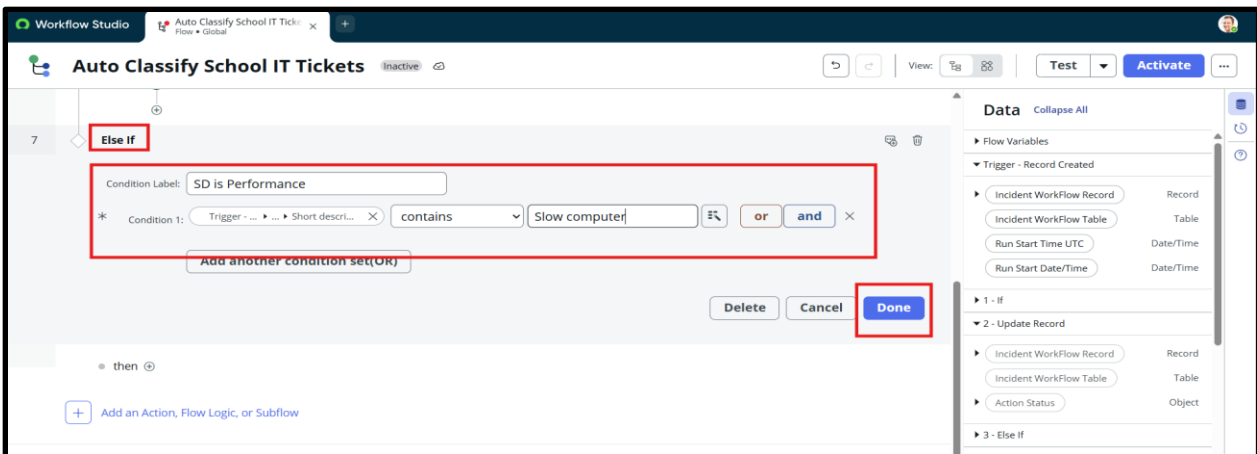
- Now again select **Else If** condition.
- Condition label : **SD is Forgot password**.
- Condition : **Short description contains Forgot password**.
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category** = **Access**.

- Set **Subcategory = Forgot Password**.
- Then Click the **Done** button.



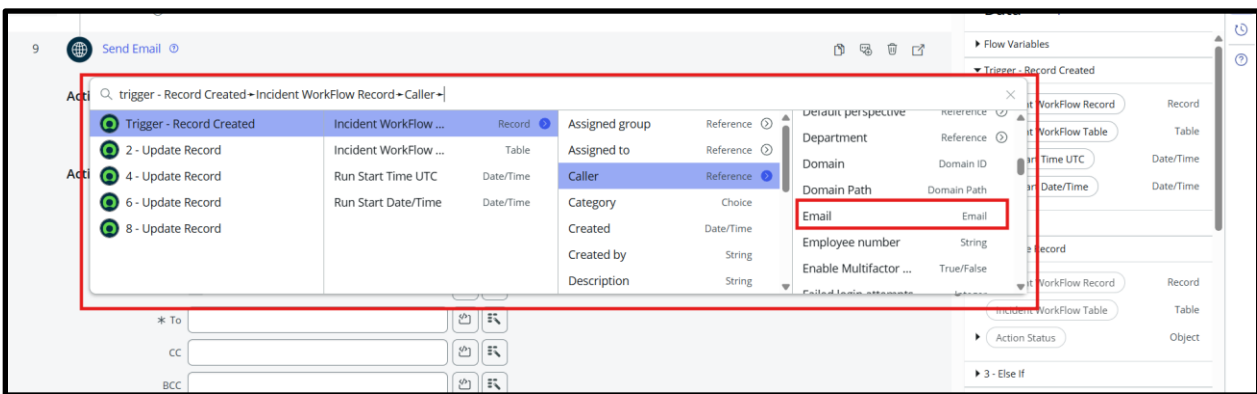
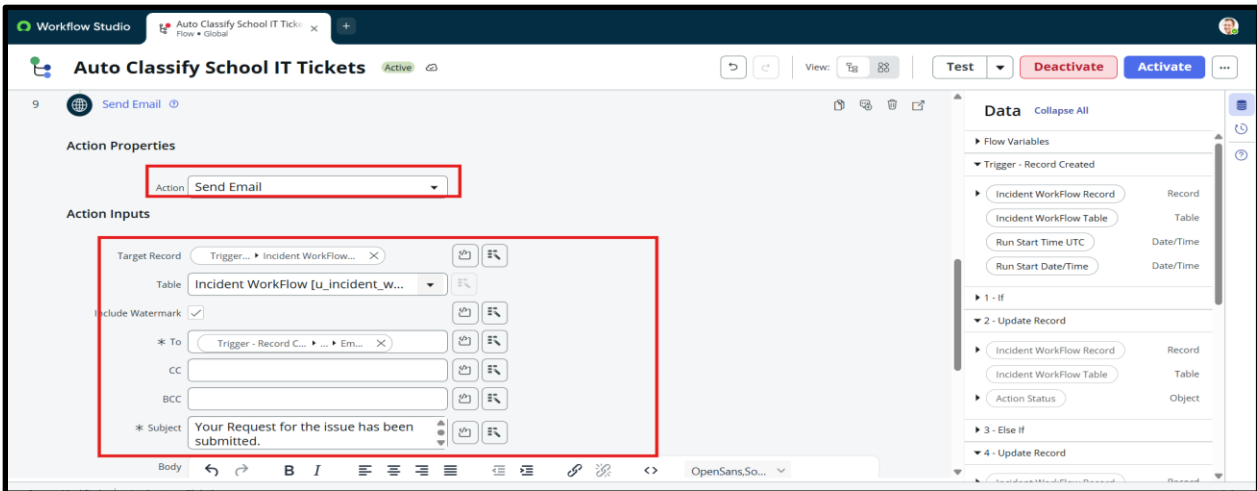
## Activity : 6 Slow Computer Classification.

- Now Again Click the **Else If** Condition.
- Condition label : **SD is Performance**.
- Condition : **Short description contains Slow Computer**.
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category = Performance**.
- Set **Subcategory = Slow computer**.
- Then Click the **Done** button.



## Activity : 7 Email Action Setup.

- After classification, an email notification is sent using Flow Designer.
- Set Action **Send Email**.
- To : Caller → Email.
- Subject : **Your Request for the issue has been submitted.**
- Body : **Ticket confirmation message.**
- Then Click the **Done** button.



- Now Click the **Activate** button from the top right corner.

## PHASE 6 : Testing, Validation & Security.

- The objective of Phase 4 is to ensure that the developed solution works as expected, meets all functional requirements, and follows basic security and access control best practices.
- This phase validates the end-to-end automation, confirms data accuracy, and ensures that only authorized users can access or modify ticket data.

## Data Validation Checks

- Mandatory fields captured correctly.
- Auto-number generated without duplication.
- Category and Subcategory values stored accurately.
- Reference fields resolve correctly to user and group tables.

### Activity : 1 Testing & Validation.

#### Test Scenario 1

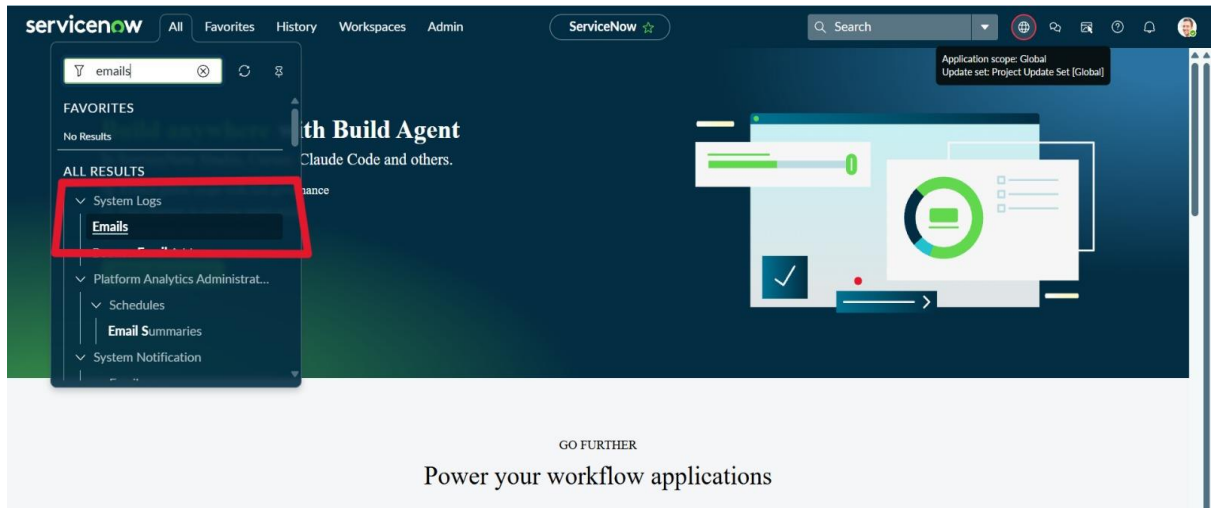
- Enter one Caller Name.
- Short Description: WiFi not working in library
- Then Save the Form.
- And Reload the Form.
- Then it fills auto as :
- Category → **Network**.
- Subcategory → **WiFi**.
- Email sent to caller Email Id.

The screenshot displays a web application interface for an 'Incident Workflow' with the ID 'INC00511'. The form contains several fields: 'Number' (auto-generated as 'INC00511'), '\* Caller' (Alfonso Griglen), 'Category' (Network), 'Subcategory' (Wi-Fi), '\* Short description' (My college Has issue with Network), and 'Description'. To the right, there are fields for 'State' (set to '-- None --'), 'Assigned group', and 'Assigned to'. At the bottom left are 'Update' and 'Delete' buttons. Red boxes highlight the 'Incident Workflow' title, the 'Number' field, the 'Caller' field, the 'Category' and 'Subcategory' dropdowns, and the 'Short description' field. A green box highlights the 'Number' field. The interface also includes a top navigation bar with icons and a bottom right corner with a small circular icon.

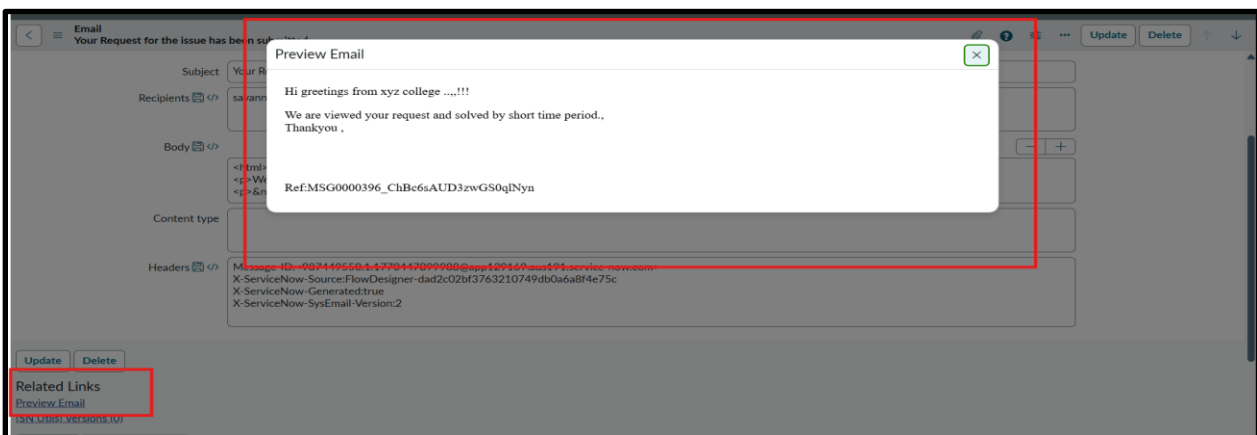
### Activity : 2 Check Email Notification.

- Open All menus → Search Emails.
- Under System logs → Select Emails.

- Search with the Given Subject Name.
- Open that Record and Under Related Links Click Preview mail Option.



| Emails Created Search |                             |                                                |            |                   |         |  |
|-----------------------|-----------------------------|------------------------------------------------|------------|-------------------|---------|--|
| Created               | Recipients                  | Subject                                        | Type       | Notification type | User ID |  |
| Search                | Search                      | *your                                          | Search     | Search            | Search  |  |
| 2026-02-06 23:04:59   | savannah.kesich@example.com | Your Request for the issue has been submitted. | send-ready | SMTP              | (empty) |  |
| 2026-02-06 23:04:52   | eliseo.wice@example.com     | Your Request for the issue has been submitted. | send-ready | SMTP              | (empty) |  |



## Activity : 3 Test Scenario 2

- Enter one Caller Name.
- Short Description: Projector not turning on.
- Then Save the Form.
- And Reload the Form.
- Then it fills auto as :
- Category → **Hardware**.
- Subcategory → **Projector**.
- Then Email sent to caller Mail id.

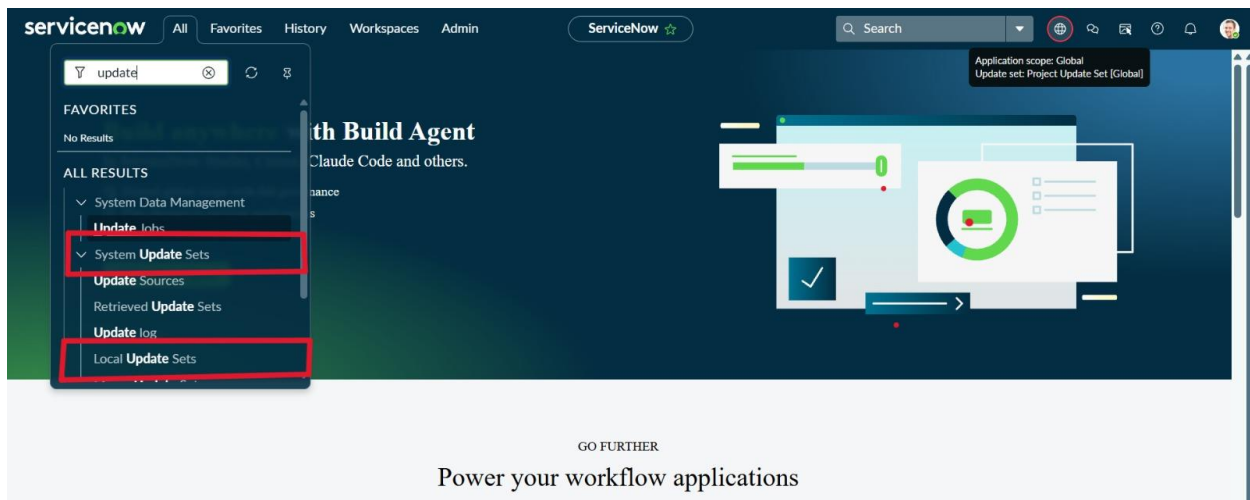
The screenshot shows a web application interface for an 'Incident Workflow'. The title bar at the top left says 'Incident Workflow' and 'INC00512'. The form contains several fields: 'Number' (INC00512), '\* Caller' (Sam Sorokin), 'Category' (Hardware), 'Subcategory' (Projector), '\* Short description' (Projector not turning on), and 'Description'. To the right of the form are fields for 'State' (set to -- None --), 'Assigned group', and 'Assigned to'. At the bottom left of the form are 'Update' and 'Delete' buttons. At the bottom right of the page is a small circular icon with a question mark.

## PHASE 5: Deployment & Final Documentation.

- The objective of Phase 4 is to deploy the developed solution to the target environment and finalize all project documentation.
- This phase ensures that the solution is stable, reusable, and ready for demonstration or production use, along with complete and clear documentation for future reference.

## Activity : 1 Make Update set to Complete State.

- Go to All menus → Search For **Local Update Sets**.
- Under **System Update sets** Select → **Local Update sets**.
- Open the **Project Update set**.
- Make the State From **In progress** to **Complete**.
- Then **Save** the Form.
- Under Related links Click **Export to XML** Option.
- Then the Xml Format the Update set data will be **downloaded** to your system.
- We can share through WhatsApp or Mail that downloaded file.
- Then the Created Update set will transform to others in this way.



The screenshot shows the 'Update Set' form for 'Project Update Set'. The 'State' dropdown is set to 'Complete', and the 'Parent' dropdown is set to 'Complete'. The 'Update' button is visible at the bottom. The form includes fields for Name, State, Parent, Release date, Install date, Installed from, and Description. The 'Application' is set to 'Global', 'Created' is '2026-02-06 20:55:31', 'Created by' is 'admin', and 'Merged to' is empty. The 'Update' button is located at the bottom left of the form.

aad36e%26sysparm\_record\_target%3Dsys\_update\_set... ☆ ⬇ S Action required ⋮

Recent download history

 **sys\_remote\_update\_set\_fdd14b82832b03546fa86f80ceaad3e8.xml**  
179 KB • Done

Back Out ⬆ ⬇

Application Global

Created 2026-09-25 10:58:29

Created by admin

Full download history ↗

Update Set

Project Update Set

✱ Name

Project Update Set

State

Complete

Parent

Release date

Install date

Installed from

Description

Application

Global

Created

2026-02-06 20:55:31

Created by

admin

Merged to

Update

Back Out

Related Links

Export to XML

Merge With Another Update Set

Scan Update Set

[SN Utils] Versions (0)

## Conclusion

- The **Auto Ticket Classification using Flow Designer** project successfully delivers a complete, end-to-end automation solution for a school IT helpdesk. The primary objective of reducing manual effort in ticket categorization and improving response efficiency has been fully achieved.
- Through structured requirement analysis, careful backend configuration, and no-code automation using Flow Designer, the system now automatically classifies IT tickets based on the short description provided by students. The implementation of dependent choice fields ensures data accuracy, while automated email notifications improve communication by instantly acknowledging ticket creation to the caller.
- The solution was developed using a phased approach, ensuring clarity, stability, and ease of maintenance. Comprehensive testing and security validation confirmed that the system functions reliably, maintains data integrity, and enforces appropriate access controls.
- Overall, this project demonstrates how ServiceNow can be effectively used to build scalable, maintainable, and user-friendly automation solutions without relying on complex scripting or machine learning. The design is well-suited for small to medium environments such as educational institutions and can be easily extended in the future to include advanced features like assignment automation, SLA tracking, or predictive intelligence.